DIGITAL EGYPT PIONEERS INITIATIVE (DEPI)

TechWise

Smart E-Commerce & Product Intelligence Platform

Requirements Specification

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Document Purpose

This document defines the stakeholder needs, user stories, use cases, functional requirements, and non-functional requirements for TechWise. It establishes a shared understanding of the system's expected behavior and provides a basis for subsequent system analysis, design, implementation, and testing.

Requirements marked as proposed or pending validation must not be treated as formally approved until the team confirms them.

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1. Introduction

1.1 Project Overview

TechWise is a Smart E-Commerce & Product Intelligence Platform designed to support users throughout the product lifecycle. Unlike a conventional e-commerce platform focused primarily on product discovery and purchasing, TechWise aims to help users understand products, evaluate alternatives, make informed purchase decisions, manage ownership, and assess upgrade or resale options.

The intended product lifecycle is:

Discover → Understand → Compare → Decide → Buy → Own → Upgrade → Resell → Transfer

TechWise combines e-commerce capabilities with structured product information, compatibility intelligence, AI-assisted shopping, deterministic purchase decision support, and a second-life marketplace governed by platform-origin and ownership eligibility rules.

1.2 Purpose of This Document

The purpose of this Requirements Specification is to:

- Identify relevant stakeholders and their needs.
- Describe representative user stories and use cases.
- Define the functional capabilities expected from TechWise.
- Establish non-functional requirements and quality expectations.
- Document essential business rules and system constraints.
- Provide requirements that can be reviewed and translated into implementation and test cases.

1.3 Scope of the Specification

This document focuses on requirements gathering. It does not claim that every described capability has already been implemented.

Detailed software architecture, database schemas, ER diagrams, class diagrams, sequence diagrams, deployment architecture, and final API contracts belong to subsequent analysis and design activities.

1.4 Terminology

Term	Definition
Product	A product model or SKU representing a type of item.

Term	Definition
Product Asset	A specific physical device or item with its own identity and lifecycle history.
Purchase	A transaction through which a product or asset is acquired.
Ownership	A particular user's period of ownership of a Product Asset.
Product Lifecycle	The sequence of product discovery, acquisition, usage, upgrade, resale, and transfer activities.
Product Truth	The structured presentation of product facts, their sources, and their confidence or verification status.
Compatibility	A determination of whether products or components can work together under defined criteria.
Purchase Decision Engine	The system capability that evaluates product candidates against user requirements and decision criteria.
Second-Life Marketplace	The marketplace for eligible previously owned products that satisfy platform-origin and ownership requirements.
Seller-Reported Information	Information supplied by a seller that has not necessarily been independently verified.
Evidence	Supporting material used to assess a claim, such as screenshots, purchase records, or product test videos.
AI-Assisted Shopping	The use of AI to interpret user intent, extract requirements, explain results, and assist product discovery and evaluation.

2. Project Problem and Objectives

2.1 Problem Statement

Online shoppers frequently face difficulty evaluating products across multiple sources. Product information may be incomplete, inconsistent, difficult to compare, or insufficient for determining whether an item meets a user's actual requirements.

Selecting a product can also require evaluating trade-offs among budget, performance, compatibility, reliability, and long-term ownership costs. Conventional product listings may not provide an integrated way to explain these trade-offs or connect the original purchase with the product's subsequent ownership and resale lifecycle.

For previously owned products, buyers may have difficulty assessing origin, condition, supporting evidence, and ownership continuity. These issues can reduce confidence in second-hand transactions.

TechWise aims to address these challenges through product intelligence, structured comparison, decision support, compatibility assessment, lifecycle continuity, and eligibility-controlled second-life trading.

2.2 Project Objectives

The objectives of TechWise are to:

1. Improve product discovery through structured search and filtering.
2. Help users understand product specifications and evaluate information quality.
3. Support product compatibility assessment using explicit criteria and available product data.
4. Assist purchase decisions by evaluating requirements, preferences, constraints, and trade-offs.
5. Preserve product identity and lifecycle continuity across ownership changes.
6. Support a second-life marketplace restricted by platform-origin and ownership eligibility rules.
7. Help users evaluate whether to buy, upgrade, or resell based on available evidence and decision criteria.
8. Use AI to interpret user intent and explain results without replacing authoritative product data or deterministic business rules.

3. Project Scope

3.1 In-Scope Capabilities

The confirmed core feature set consists of the following six capabilities.

3.1.1 AI Shopping Intelligence

Supports natural-language shopping requests, interpretation of user needs, extraction of requirements, and understandable explanations of relevant product options.

3.1.2 Verified Second-Life Marketplace

Supports the discovery and listing of eligible previously owned products, subject to platform-origin, current ownership, evidence, and other applicable eligibility checks.

The term “verified” refers to the platform's eligibility and evidence assessment process. It must not be interpreted as proof of physical inspection or independent verification unless such a process is actually implemented.

3.1.3 Product Compatibility Graph

Represents compatibility relationships between products and components and supports compatibility assessments based on available structured facts and defined rules.

3.1.4 Purchase Decision Engine

Evaluates eligible product candidates against user requirements, mandatory constraints, preferences, and relevant trade-offs.

3.1.5 Product Truth Layer

Organizes product information, distinguishes factual records from seller-reported claims, and presents relevant sources or supporting evidence and confidence information where available.

3.1.6 Buy vs Upgrade vs Resell Engine

Supports lifecycle-related decisions by comparing available options against user requirements, product condition, relevant costs, and lifecycle information.

3.2 Cross-Cutting Capability: Advanced Smart Filtering

Advanced Smart Filtering is a cross-cutting system capability rather than a separate core feature.

It supports relevant product discovery and evaluation by applying structured criteria, mandatory constraints, and user-selected preferences.

3.3 Out-of-Scope and Unconfirmed Items

The following must not be presented as confirmed requirements unless separately approved:

- A specific payment provider or payment-processing integration.
- A specific AI model, AI provider, orchestration framework, or hosting provider.
- A specific backend framework, frontend framework, database engine, or deployment architecture.
- Physical inspection of every second-life product by the platform.
- Automatic verification of every seller claim.
- Guaranteed accuracy of AI-generated explanations or recommendations.
- Specific escrow, refund, or dispute-resolution mechanisms not yet approved as business rules.
- Any production deployment, performance benchmark, or security certification that has not been demonstrated.

These items may be considered during later design and planning if the team approves them.

4. Stakeholder Analysis

The stakeholder roles below represent the proposed analysis of parties relevant to TechWise. Candidate roles and their responsibilities must be validated by the team before being treated as final authorization rules.

Stakeholder	Main Needs	Expected System Support
Prospective Shopper	Find relevant products and understand available options.	Search, filtering, product details, and AI-assisted discovery.
Registered Shopper	Compare candidates and make informed purchase decisions.	Comparison, compatibility assessment, decision explanations, and saved or managed user information where approved.
Product Owner	Track a specific physical product and its ownership period.	Product Asset information, ownership records, and lifecycle history subject to authorization.
Second-Life Seller	Determine resale eligibility and provide listing information and evidence.	Eligibility checks, evidence submission, listing status, and listing management.
Second-Life Buyer	Evaluate a previously owned product before acquiring it.	Product condition information, evidence presentation, provenance information, and available ownership-transfer status.
Platform Administrator or Reviewer	Manage platform records and review cases within authorized permissions.	Administrative functions, eligibility review, and controlled record management if included in the approved scope.
Falcon Team	Convert requirements into an implementable and testable system.	Clear requirements, priorities, business rules, and acceptance criteria.
Project Supervisor / DEPI Evaluator	Assess scope, documentation quality, technical reasoning, and project progress.	Traceable requirements and consistent project documentation.

4.1 Stakeholder Priorities

The primary user-facing priorities are:

- Improve the relevance and clarity of product information.
- Help users evaluate compatibility and purchase trade-offs.
- Preserve product identity and ownership continuity.
- Reduce uncertainty around second-life product eligibility and supporting evidence.
- Maintain secure access to ownership and personal information.

4.2 Stakeholder Validation

The team should validate the proposed user roles, permissions, and stakeholder priorities before implementation. Any unresolved decisions must be recorded rather than silently treated as approved.

5. User Stories

User stories describe representative needs from the perspective of a user. They are intended to guide requirement validation and later acceptance testing.

US-01 — Natural-Language Product Discovery

As a shopper, I want to describe what I need in natural language so that TechWise can help me identify relevant products without requiring me to know every technical specification.

Acceptance criteria:

- The system accepts a natural-language shopping request.
- Interpreted requirements can be reviewed or clarified when necessary.
- Product results are based on available product records and supported search criteria.
- The system communicates when information is missing or insufficient.

US-02 — Structured Product Filtering

As a shopper, I want to filter products by relevant specifications and constraints so that I can narrow the available options.

Acceptance criteria:

- Supported filters can be applied to available product data.
- Mandatory criteria are distinguished from optional preferences where applicable.
- The system does not present a product as satisfying a criterion when the required fact is unavailable.

US-03 — Product Comparison

As a shopper, I want to compare multiple product candidates using consistent criteria so that I can understand their differences.

Acceptance criteria:

- Comparable attributes are presented consistently.
- Missing values are identified instead of silently treated as equivalent.
- The user can identify important differences and trade-offs.

US-04 — Compatibility Assessment

As a shopper, I want to check whether a product or component is compatible with another item so that I can avoid unsuitable purchases.

Acceptance criteria:

- Compatibility is evaluated against available structured facts and applicable rules.
- The result distinguishes compatible, incompatible, and unknown or insufficient-data outcomes when relevant.
- Explanations identify the available facts or constraints that support the result.

US-05 — Purchase Decision Support

As a shopper, I want TechWise to evaluate product candidates against my requirements and priorities so that I can make a more informed purchase decision.

Acceptance criteria:

- Mandatory constraints are evaluated separately from weighted preferences.
- The system explains important trade-offs.
- A candidate that violates a mandatory constraint is not described as fully satisfying the request.
- The final decision remains understandable to the user.

US-06 — Product Information Assessment

As a shopper, I want to understand the source and reliability of relevant product information so that I can distinguish documented facts from unverified claims.

Acceptance criteria:

- Available product information is presented with appropriate source or evidence context.
- Seller-reported claims are not automatically described as independently verified facts.

- Unknown or conflicting information is communicated where relevant.

US-07 — Ownership Lifecycle Tracking

As a product owner, I want a product's lifecycle history to remain associated with the same physical asset so that ownership changes do not erase the item's earlier history.

Acceptance criteria:

- A Product Asset remains distinguishable from its product model.
- Ownership periods are represented separately from the asset's identity.
- Authorized users can access relevant lifecycle information.
- Previous owners' personal information is not exposed to unauthorized users.

US-08 — Resale Eligibility

As a current owner, I want TechWise to assess whether my product is eligible for second-life listing so that I understand the platform's requirements before submitting it.

Acceptance criteria:

- Eligibility is evaluated using the applicable platform-origin and ownership rules.
- Required evidence is identified.
- Ineligible products are not treated as eligible merely because the seller submits a listing.
- The outcome can identify missing requirements or evidence.

US-09 — Second-Life Listing Evaluation

As a potential buyer, I want to review a second-life listing and its available supporting evidence so that I can assess the product before deciding whether to buy it.

Acceptance criteria:

- The listing presents available product and condition information.
- Seller-reported statements remain distinguishable from supported evidence.
- Missing evidence or unresolved eligibility information is not hidden.
- The system does not claim a physical inspection occurred unless it actually did.

US-10 — Ownership Transfer

As a second-life buyer, I want a successful ownership transfer to update the current ownership record while preserving the product's earlier lifecycle history.

Acceptance criteria:

- The Product Asset retains its original identity.
- The current ownership period is updated through an authorized process.
- Earlier ownership records remain part of the lifecycle history.
- The new owner's ownership code or reference follows the approved business rules.
- The original first-purchase date is not reset by resale.

US-11 — Buy, Upgrade, or Resell Evaluation

As a product owner, I want to compare buying a replacement, upgrading my current product, or reselling it so that I can evaluate the available lifecycle options.

Acceptance criteria:

- The system compares options using the available product, cost, condition, and requirement data.
- Important assumptions and missing information are disclosed.
- Calculations are based on defined inputs and deterministic rules.
- The explanation does not present an unsupported estimate as a guaranteed outcome.

US-12 — Authorized Administration

As an authorized platform administrator or reviewer, I want to manage permitted records and review cases so that platform rules can be applied consistently.

Acceptance criteria:

- Administrative capabilities are limited to explicitly authorized roles.
- Sensitive ownership and personal information is protected.
- Important status changes can be traced where the approved design requires audit records.

Status: *The administrator role and its detailed permissions require team validation.*

6. Use Cases

The following use cases describe major user-system interactions. They are requirements-level descriptions, not final sequence diagrams or UI specifications.

UC-01 — Discover Products

Primary actor: Shopper

Preconditions:

- Product information is available to the search process.

- Any required access conditions have been satisfied.

Main flow:

1. The shopper opens product discovery.
2. The shopper enters a query or selects supported filters.
3. The system interprets the request and applies available search criteria.
4. The system returns relevant product candidates.
5. The shopper reviews the results and opens a product.

Alternative flows:

- If no results match, the system explains that no matching candidates were found and may suggest relaxing optional criteria.
- If required information is missing, the system requests clarification or indicates the limitation.

Postcondition: The shopper receives a result set or an informative empty state.

UC-02 — Compare Products

Primary actor: Shopper

Preconditions:

- At least two eligible product candidates are available for comparison.

Main flow:

1. The shopper selects products.
2. The system retrieves comparable attributes.
3. The system presents a comparison.
4. The shopper reviews similarities, differences, and missing information.
5. The shopper proceeds to a decision or returns to discovery.

Alternative flows:

- If a value is unavailable for one candidate, the system identifies it as unavailable rather than inventing a value.

Postcondition: The shopper can inspect a consistent comparison.

UC-03 — Check Compatibility

Primary actor: Shopper

Preconditions:

- Relevant product or component records and compatibility rules are available.

Main flow:

1. The shopper selects the items to evaluate.

2. The system retrieves their relevant specifications.
3. The compatibility service evaluates applicable constraints.
4. The system presents the result and its supporting reasoning.
5. The shopper reviews any limitations or alternatives.

Alternative flows:

- If required specifications are missing, the result is unknown or insufficient data.
- If a mandatory compatibility condition fails, the system identifies the failed condition.

Postcondition: A compatibility outcome is presented without confusing missing data with confirmed compatibility.

UC-04 — Evaluate a Purchase Decision

Primary actor: Shopper

Preconditions:

- Candidate products and the necessary decision inputs are available.

Main flow:

1. The shopper supplies requirements, constraints, and preferences.
2. The system interprets the request into structured criteria.
3. The system validates the criteria.
4. The decision engine evaluates candidates against mandatory constraints and preferences.
5. The system presents rankings, trade-offs, and relevant explanations.
6. The shopper reviews the outcome and chooses a next action.

Alternative flows:

- If criteria conflict, the system requests clarification or explains the conflict.
- If data is insufficient, the system discloses the limitation.
- If no candidate satisfies all mandatory constraints, the system must not label a violating candidate as a full match.

Postcondition: The shopper receives a decision-support result and understands its main limitations.

UC-05 — Inspect Product Information

Primary actor: Shopper

Preconditions:

- A product record exists.

Main flow:

1. The shopper opens a product's details.

2. The system presents available specifications and relevant product facts.
3. The system presents source, evidence, or confidence information where available.
4. The shopper reviews the information.
5. The shopper proceeds to comparison or decision support.

Alternative flows:

- Conflicting, incomplete, or seller-reported information is appropriately identified.

Postcondition: The shopper can distinguish available facts from claims and unknowns.

UC-06 — View an Owned Product's Lifecycle

Primary actor: Authorized product owner

Preconditions:

- The Product Asset and an authorized ownership record exist.

Main flow:

1. The owner opens the owned-products area.
2. The system checks access permissions.
3. The system retrieves the relevant Product Asset and ownership period.
4. The system presents authorized lifecycle events.
5. The owner reviews available history and relevant actions.

Alternative flows:

- Unauthorized requests are denied without disclosing another user's private information.

Postcondition: The authorized owner can view permitted information about the asset and their ownership period.

UC-07 — Submit a Second-Life Listing

Primary actor: Current product owner acting as seller

Preconditions:

- The seller can establish current ownership through the approved process.
- The product meets the platform-origin requirements.
- The Product Asset is eligible for evaluation.

Main flow:

1. The seller selects a Product Asset.
2. The system evaluates initial eligibility.
3. The system identifies required listing information and evidence.
4. The seller submits the listing and required evidence.
5. The system validates the submission.

6. The listing receives an appropriate status based on the result.

Alternative flows:

- If ownership or origin requirements fail, the listing is rejected or blocked according to the approved rules.
- If evidence is missing, the system requests the missing information.
- If a blocking dispute exists, eligibility is handled according to the approved dispute rules.

Postcondition: The listing is recorded with an eligibility or review status. Submission alone does not imply that the product has been physically inspected.

UC-08 — Review a Second-Life Listing

Primary actor: Prospective second-life buyer

Preconditions:

- A listing is available for viewing.

Main flow:

1. The buyer opens the listing.
2. The system presents product identity, condition information, price, and available evidence.
3. The buyer reviews the listing's status and supporting information.
4. The buyer decides whether to continue with the applicable purchase process.

Alternative flows:

- Missing or unverified information is clearly identified.

Postcondition: The buyer can assess the available information before proceeding.

UC-09 — Transfer Ownership

Primary actor: Authorized current owner and eligible new owner

Preconditions:

- The transfer is permitted by the approved business rules.
- Required transaction and eligibility conditions have been satisfied.

Main flow:

1. The system identifies the Product Asset involved.
2. The system validates the transfer conditions.
3. The authorized transfer process is completed.
4. The previous ownership period is preserved as historical data.
5. A new ownership period is recorded.
6. The product's original lifecycle and first-purchase date remain unchanged.

Alternative flows:

- If validation fails, the transfer is not completed.
- If a blocking dispute exists, the system follows the approved dispute process.

Postcondition: Ownership changes without resetting the physical asset's age or erasing historical ownership.

UC-10 — Evaluate Buy vs Upgrade vs Resell

Primary actor: Product owner

Preconditions:

- Relevant current-product information and enough data for the requested evaluation are available.

Main flow:

1. The owner selects a product or Product Asset.
2. The system collects available condition, cost, and requirement information.
3. The system evaluates applicable buy, upgrade, and resale options.
4. The system compares the options using defined calculations and assumptions.
5. The system presents the results and limitations.
6. The owner reviews the recommendation and chooses a next action.

Alternative flows:

- If a reliable comparison cannot be calculated because required inputs are missing, the system identifies those inputs and limits the conclusion.

Postcondition: The owner receives a transparent comparison of available lifecycle choices.

7. Functional Requirements

7.1 Requirement Classification

The following priority labels are used:

- **Must:** Necessary to satisfy the defined core capability or an essential business rule.
- **Should:** Important to the intended user experience but subject to implementation planning.
- **Could:** A useful enhancement if time and resources permit.

The priorities below are proposed for team validation. They are not a claim that DEPI has assigned these priorities.

7.2 AI Shopping Intelligence

FR-AI-01 — Natural-Language Input

The system shall accept shopping requests expressed in natural language.

Priority: *Must*

FR-AI-02 — Structured Requirement Extraction

The system shall convert interpretable shopping requests into structured requirements, preferences, and constraints before downstream evaluation.

Priority: *Must*

FR-AI-03 — Clarification of Ambiguous Requests

The system should request clarification when ambiguity materially affects the interpretation or evaluation of a shopping request.

Priority: *Should*

FR-AI-04 — Recommendation Explanation

The system shall provide understandable explanations for relevant recommendations using available product facts and decision results.

Priority: *Must*

FR-AI-05 — AI Boundary

AI-generated interpretation or explanation shall not independently override product records, authorization checks, compatibility rules, ownership state, or deterministic business decisions.

Priority: *Must*

FR-AI-06 — Uncertainty Disclosure

The system should communicate when a recommendation depends on incomplete data, ambiguous intent, or unsupported assumptions.

Priority: *Must*

7.3 Product Discovery and Smart Filtering

FR-SF-01 — Product Search

The system shall allow shoppers to search available product records using supported query criteria.

Priority: *Must*

FR-SF-02 — Structured Filtering

The system shall apply supported filters to available product attributes.

Priority: *Must*

FR-SF-03 — Mandatory Constraints

Where the user defines mandatory criteria, the system shall distinguish them from optional preferences.

Priority: *Must*

FR-SF-04 — Filter Combination

The system should support combining multiple supported filters.

Priority: *Should*

FR-SF-05 — Empty Results

The system shall communicate when no candidates satisfy the applied criteria.

Priority: *Must*

FR-SF-06 — Missing Attribute Handling

The system shall not treat an unknown attribute as proof that a product satisfies a filter.

Priority: *Must*

7.4 Product Comparison

FR-PC-01 — Attribute Comparison

The system shall present selected products using consistent comparison attributes when comparable data is available.

Priority: *Must*

FR-PC-02 — Missing Data Identification

The system shall identify unavailable comparison values.

Priority: *Must*

FR-PC-03 — Difference Explanation

The system should highlight relevant differences that affect the user's stated requirements.

Priority: *Should*

FR-PC-04 — Evidence-Aware Comparison

Where relevant evidence or source information is available, the system should expose that context for compared product facts.

Priority: *Should*

7.5 Product Compatibility Graph

FR-CG-01 — Compatibility Data Representation

The system shall represent supported compatibility relationships using structured product or component data.

Priority: *Must*

FR-CG-02 — Rule-Based Evaluation

The system shall evaluate compatibility using defined criteria and available structured facts.

Priority: *Must*

FR-CG-03 — Explicit Outcomes

The system shall distinguish supported compatible and incompatible outcomes from unknown or insufficient-data outcomes.

Priority: *Must*

FR-CG-04 — Reasoning Disclosure

The system should identify the relevant facts or constraints that support a compatibility result.

Priority: *Should*

FR-CG-05 — AI Boundary for Compatibility

AI may help interpret the user's question or explain a result, but shall not independently establish compatibility in place of the applicable rules and product data.

Priority: *Must*

7.6 Purchase Decision Engine

FR-PD-01 — Decision Input Collection

The system shall support the collection of user requirements, mandatory constraints, and preferences relevant to a purchase decision.

Priority: *Must*

FR-PD-02 — Candidate Evaluation

The system shall evaluate available candidates against the defined decision criteria.

Priority: *Must*

FR-PD-03 — Constraint Validation

The system shall evaluate mandatory constraints separately from optional preferences.

Priority: *Must*

FR-PD-04 — Ranking

The system shall rank eligible candidates using defined, reproducible evaluation rules.

Priority: *Must*

FR-PD-05 — Trade-Off Explanation

The system should explain material trade-offs between candidates.

Priority: *Should*

FR-PD-06 — No False Full Match

The system shall not label a candidate as fully satisfying the request when the candidate violates a mandatory constraint.

Priority: *Must*

FR-PD-07 — Insufficient Data

The system shall communicate when insufficient information prevents a reliable evaluation.

Priority: *Must*

FR-PD-08 — Decision Transparency

Where a result depends on assumptions, weights, or estimates, the system should disclose the relevant assumptions.

Priority: *Should*

7.7 Product Truth Layer

FR-PT-01 — Structured Product Facts

The system shall organize supported product attributes as structured data.

Priority: *Must*

FR-PT-02 — Source Context

Where source information is available, the system should associate relevant product facts with their sources or evidence.

Priority: *Should*

FR-PT-03 — Seller-Reported Claims

The system shall distinguish seller-reported information from information that has supporting evidence or an explicit platform assessment.

Priority: *Must*

FR-PT-04 — Evidence and Confidence

The system should present available evidence and platform confidence or assessment information without implying certainty beyond what the evidence supports.

Priority: *Should*

FR-PT-05 — Conflicting Information

The system should identify material conflicts or gaps in product information when they can be detected.

Priority: *Should*

FR-PT-06 — No Fabricated Facts

The system shall not present AI-generated or inferred values as authoritative product facts without the required data and validation.

Priority: *Must*

7.8 Verified Second-Life Marketplace

FR-SL-01 — Eligible Listing Discovery

The system shall allow users to discover second-life listings that satisfy the marketplace's applicable visibility and eligibility rules.

Priority: *Must*

FR-SL-02 — Platform-Origin Requirement

A product shall satisfy the applicable TechWise platform-purchase or origin requirement before becoming eligible for second-life listing.

Priority: *Must*

FR-SL-03 — Current Ownership Validation

The system shall validate the seller's current ownership through the approved ownership and purchase-history process.

Priority: *Must*

FR-SL-04 — Product Asset Association

A second-life listing shall be associated with the relevant Product Asset rather than treating the listing as an unrelated new product.

Priority: *Must*

FR-SL-05 — Evidence Submission

The system shall support submission of the evidence required by the approved listing rules. For electronic products, the current project direction includes screenshots and a product test video.

Priority: *Must*

FR-SL-06 — Eligibility Result

The system shall evaluate applicable listing conditions, including ownership, origin, listing state, evidence requirements, and blocking disputes where those rules are defined.

Priority: *Must*

FR-SL-07 — Listing Status

The system shall represent the listing's applicable status and prevent an ineligible listing from being presented as an eligible active listing.

Priority: *Must*

FR-SL-08 — Evidence-Aware Presentation

The system shall present relevant listing evidence and seller-reported condition information in a way that does not imply independent physical verification unless it has occurred.

Priority: *Must*

FR-SL-09 — Rejection Feedback

The system should explain missing eligibility requirements or evidence when a listing cannot proceed.

Priority: *Should*

FR-SL-10 — Authorized Listing Management

The system should allow an authorized seller to review and manage permitted listing information and status.

Priority: *Should*

7.9 Product Lifecycle and Ownership

FR-PL-01 — Separate Product and Product Asset

The system shall distinguish a product model or SKU from a specific physical Product Asset.

Priority: *Must*

FR-PL-02 — Ownership Periods

The system shall represent ownership separately from the Product Asset's identity.

Priority: *Must*

FR-PL-03 — Lifecycle Continuity

The Product Asset's identity and historical lifecycle shall remain associated with the same physical item across resale and ownership transfer.

Priority: *Must*

FR-PL-04 — Original First-Purchase Date

The system shall preserve the original first-purchase date after resale.

Priority: *Must*

FR-PL-05 — Historical Ownership

The system shall preserve previous ownership records as historical records when ownership changes.

Priority: *Must*

FR-PL-06 — Ownership Reference Rules

The system shall apply the approved ownership reference or code rules. Under the current project direction, the previous owner's code becomes invalid for the new ownership period, and the new owner receives a new code while the original first-purchase date remains unchanged.

Priority: *Must*

FR-PL-07 — Privacy of Previous Owners

The system shall not expose previous owners' personal information to unauthorized users.

Priority: *Must*

FR-PL-08 — Transfer Validation

The system shall validate authorization and applicable transfer conditions before recording an ownership change.

Priority: *Must*

7.10 Buy vs Upgrade vs Resell Engine

FR-BUR-01 — Lifecycle Option Comparison

The system shall support evaluation of applicable buy, upgrade, and resell options using available data.

Priority: *Must*

FR-BUR-02 — Requirement-Based Evaluation

The system should evaluate options against the user's current requirements and constraints.

Priority: *Should*

FR-BUR-03 — Cost and Condition Inputs

The system shall use available relevant cost, product-condition, and lifecycle information in applicable evaluations.

Priority: *Must*

FR-BUR-04 — Deterministic Calculations

Calculations used for the decision shall be performed using defined rules and validated inputs rather than relying solely on AI-generated numerical conclusions.

Priority: *Must*

FR-BUR-05 — Assumption Disclosure

The system should disclose important assumptions, missing inputs, or estimates that affect the comparison.

Priority: *Should*

FR-BUR-06 — Explainable Outcome

The system should explain the main factors influencing the comparison.

Priority: *Should*

7.11 Access Control and Administrative Requirements

FR-AC-01 — Authorization

The system shall restrict protected operations to authorized users and roles.

Priority: *Must*

FR-AC-02 — Ownership Access

The system shall restrict access to ownership and lifecycle information according to approved permissions.

Priority: *Must*

FR-AC-03 — Eligibility Enforcement

The system shall enforce listing and ownership rules on the backend rather than relying only on frontend controls or AI output.

Priority: *Must*

FR-AC-04 — Status Change Traceability

Where required by the approved business process, important ownership and listing status changes should be recorded for traceability.

Priority: *Should*

Detailed role definitions and permission matrices must be finalized during requirements validation and system design.

8. Non-Functional Requirements

Non-functional requirements define quality expectations and constraints. The requirements below establish intended quality attributes. Numeric performance targets, availability targets,

and testing thresholds remain proposed until validated against the project scope and available infrastructure.

8.1 Performance

NFR-PER-01 —

Common user interactions should return feedback within an acceptable time under the agreed test environment.

Priority: *Proposed*

NFR-PER-02 —

Product search and filtering should remain responsive for the dataset and load defined by the project test plan.

Priority: *Proposed*

NFR-PER-03 —

Purchase decision and compatibility evaluations should complete within agreed test thresholds for representative inputs.

Priority: *Proposed*

NFR-PER-04 —

Performance claims shall be supported by measurements under documented conditions rather than assumed from the design.

Priority: *Proposed*

8.2 Security and Privacy

NFR-SEC-01 —

Protected information and operations shall require appropriate authentication and authorization according to the approved access model.

Priority: *Must*

NFR-SEC-02 —

Personal, purchase, and ownership information shall be protected against unauthorized access.

Priority: *Must*

NFR-SEC-03 —

Critical business rules and permissions shall be enforced by trusted backend logic, not solely by client-side controls.

Priority: *Must*

NFR-SEC-04 —

User inputs and submitted evidence metadata shall be validated before being processed or persisted.

Priority: *Must*

NFR-SEC-05 —

The system shall prevent unauthorized disclosure of previous owners' personal information.

Priority: *Must*

NFR-SEC-06 —

Error messages shall provide useful feedback without exposing credentials, sensitive records, or internal implementation details.

Priority: *Must*

8.3 Reliability and Data Integrity

NFR-REL-01 —

Ownership changes shall not reset the Product Asset's original first-purchase date or erase historical ownership records.

Priority: *Must*

NFR-REL-02 —

The system shall maintain consistent relationships among products, Product Assets, purchases, ownership periods, and second-life listings.

Priority: *Must*

NFR-REL-03 —

Failed validation or processing shall produce an appropriate error or status without falsely reporting success.

Priority: *Must*

NFR-REL-04 —

The system should handle repeated or conflicting operations in a way that prevents inconsistent ownership or listing states.

Priority: *Must*

8.4 Usability and Accessibility

NFR-USA-01 —

The system should explain technical product information and decision outcomes in language appropriate for the intended user.

Priority: *Should*

NFR-USA-02 —

Common actions, feedback messages, and navigation patterns should behave consistently across supported screens.

Priority: *Should*

NFR-USA-03 —

The user interface should remain usable across the desktop and mobile screen sizes included in the approved scope.

Priority: *Should*

NFR-USA-04 —

The system should explain validation errors and provide a clear way to correct user input where possible.

Priority: *Should*

NFR-USA-05 —

Recommendations, compatibility results, and eligibility outcomes should communicate their main reasons and limitations.

Priority: *Should*

8.5 Maintainability

NFR-MNT-01 —

AI interpretation, deterministic business logic, data access, and user-interface responsibilities should be separated sufficiently to support testing and maintenance.

Priority: *Should*

NFR-MNT-02 —

Core business rules should be testable independently of AI-generated explanations where technically appropriate.

Priority: *Should*

NFR-MNT-03 —

Implementation and documentation shall use the defined distinctions among Product, Product Asset, Purchase, and Ownership.

Priority: *Should*

NFR-MNT-04 —

Requirements and related implementation changes should be traceable through the team's agreed development workflow.

Priority: *Should*

8.6 Data Quality and Explainability

NFR-DQ-01 —

Where source information is available, relevant product facts should retain sufficient source context for appropriate review.

Priority: *Must*

NFR-DQ-02 —

Unknown or unavailable data shall not be silently converted into a positive claim.

Priority: *Must*

NFR-DQ-03 —

Deterministic calculations and rule-based evaluations should produce consistent results for the same validated inputs and rule version.

Priority: *Must*

NFR-DQ-04 —

AI-generated explanations shall remain consistent with the underlying facts and evaluation results.

Priority: *Must*

NFR-DQ-05 —

The system shall not describe submitted evidence as proof of a fact beyond what the evidence and review process actually establish.

Priority: *Must*

8.7 Scalability and Availability

NFR-SCA-01 —

The system design should allow the product catalogue and lifecycle records to grow without requiring unnecessary redesign of core business concepts.

Priority: *Proposed*

NFR-AVL-01 —

Availability requirements and recovery targets shall be defined according to the approved deployment environment and project scope.

Priority: *Proposed*

No uptime percentage or recovery-time guarantee is claimed until a deployment environment and measurement method are confirmed.

9. Business Rules and Constraints

The following rules reflect the current project direction and must remain consistent across requirements, design, implementation, and testing.

9.1 AI and Source-of-Truth Rules

BR-01: AI is not the authoritative source of product facts or database state.

BR-02: AI may interpret intent, extract requirements, explain results, summarize information, and support review intelligence.

BR-03: Validation, authorization, eligibility, compatibility evaluation, ranking calculations, ownership state, and other critical business rules must be enforced by deterministic backend logic or other explicitly defined authoritative systems.

BR-04: AI explanations must not override a validated business result.

9.2 Product and Ownership Rules

BR-05: A Product is distinct from a Product Asset.

BR-06: A Product Asset represents a specific physical item and retains its identity across ownership changes.

BR-07: Ownership represents a particular user's ownership period and is not the same as the Product Asset.

BR-08: Product age and original first-purchase date do not reset after resale.

BR-09: Previous ownership records remain part of the lifecycle history, while previous owners' personal information remains protected.

9.3 Second-Life Rules

BR-10: Second-life listing eligibility depends on the applicable TechWise platform-origin or purchase-history requirements.

BR-11: The current seller must satisfy the approved current-ownership validation process.

BR-12: A listing must be associated with the relevant Product Asset.

BR-13: Required evidence must be supplied and evaluated according to the approved listing rules. For electronics, the current direction includes screenshots and a product test video.

BR-14: Eligibility may depend on listing state, evidence completeness, current ownership, platform origin, and blocking disputes as defined by the approved business process.

BR-15: Seller-reported claims, submitted evidence, and platform confidence or assessment must remain distinguishable.

BR-16: No physical platform inspection or verification shall be claimed unless that activity is genuinely performed by an implemented process.

BR-17: After an approved resale, the previous ownership code becomes invalid for the new ownership period and the new owner receives a new code, while the original first-purchase date and asset history remain intact.

10. External Interface and Data Requirements

This section identifies interface and data needs at the requirements level. It does not prescribe an unconfirmed technology stack or final API design.

10.1 User Interfaces

The system should provide interfaces appropriate to its approved user-facing scope for:

- Product discovery and search.
- Product details and structured comparison.
- AI-assisted shopping and decision support.
- Compatibility assessment.
- Product truth and evidence presentation.
- Owned-product and lifecycle views.
- Second-life marketplace discovery and listing management.
- Ownership and eligibility-related actions.

The exact screens, routes, and visual specifications are maintained in the project's design deliverables.

10.2 External Interfaces

Potential external interfaces may include product information sources, AI services, purchase or payment services, and other integrations. Their inclusion, providers, data contracts, and availability must be confirmed by the team before being treated as committed dependencies.

This document does not assert that a particular external service or provider has been selected.

10.3 Conceptual Data Requirements

At the requirements level, TechWise must preserve the distinction among:

- Product
- Product Asset
- Purchase
- Ownership
- Product facts and supporting evidence
- Compatibility relationships
- Decision criteria and evaluation results
- Second-life listing and eligibility status
- Lifecycle and ownership-transfer history

The exact entities, attributes, keys, relationships, and database constraints are to be specified in the System Analysis & Design phase.

11. Acceptance and Validation Approach

Requirements must be reviewed for clarity, consistency, feasibility, and testability before implementation.

11.1 Validation Principles

A requirement should:

1. Describe one clear expected behavior or quality expectation.
2. Be consistent with the approved project scope.
3. Avoid prescribing an unconfirmed technology without justification.
4. Be verifiable through inspection, analysis, demonstration, or testing.
5. Identify relevant failure conditions or missing-data behavior.
6. Preserve the boundary between AI assistance and authoritative backend decisions.

11.2 Representative Acceptance Scenarios

ID	Scenario	Expected Result
AT-01	A search request has no matching products.	The system communicates the empty result and does not invent products.
AT-02	A product lacks a required specification.	The missing value is identified and is not treated as a confirmed match.
AT-03	A candidate violates a mandatory decision constraint.	The candidate is not labelled as a full match.
AT-04	Compatibility cannot be determined because data is missing.	The result is identified as unknown or insufficient data.
AT-05	A seller submits a listing without satisfying origin or ownership requirements.	The listing is blocked or handled according to the eligibility rules.
AT-06	A second-life listing contains seller-reported condition claims.	The claims are distinguishable from supporting evidence and platform assessment.
AT-07	Ownership changes after resale.	The Product Asset and original first-purchase date remain unchanged.
AT-08	An unauthorized user requests private ownership information.	Access is denied without disclosing the protected information.
AT-09	An AI-generated explanation conflicts with the validated backend result.	The validated result remains authoritative.
AT-10	A buy-versus-upgrade comparison lacks a required cost input.	The system identifies the limitation rather than presenting an unsupported definitive conclusion.

11.3 Requirements Review

Before implementation begins, the team should review the functional requirements, stakeholder needs, user stories, use cases, and non-functional requirements together.

The review should identify:

- Conflicting or ambiguous requirements.
- Requirements that depend on unconfirmed integrations.

- Missing business rules.
- Requirements without a practical verification method.
- Proposed targets that need measurable thresholds.
- Requirements that should be deferred or excluded from the approved scope.

12. Requirements Traceability

Traceability connects stakeholder needs to system capabilities and later testing.

Stakeholder Need	Related Capability	Requirement Group	Validation Example
Find relevant products	AI Shopping Intelligence and Smart Filtering	FR-AI, FR-SF	AT-01, AT-02
Compare alternatives	Product Comparison	FR-PC	AT-02
Avoid incompatible purchases	Product Compatibility Graph	FR-CG	AT-04
Make informed purchase decisions	Purchase Decision Engine	FR-PD	AT-03, AT-09
Understand product information	Product Truth Layer	FR-PT	AT-06
Assess resale eligibility	Verified Second-Life Marketplace	FR-SL	AT-05, AT-06
Preserve lifecycle continuity	Ownership and Product Lifecycle	FR-PL	AT-07, AT-08
Compare lifecycle choices	Buy vs Upgrade vs Resell Engine	FR-BUR	AT-10
Protect private information	Access Control and Security	FR-AC, NFR-SEC	AT-08

A detailed traceability matrix linking each approved requirement to a design artifact, implementation task, and test case should be maintained during subsequent project phases.

13. Assumptions, Open Decisions, and Approval

13.1 Confirmed Project Direction

The following are established elements of the current TechWise project direction:

- Official project name and description.
- The six core capabilities listed in Section 3.
- Advanced Smart Filtering as a cross-cutting capability.
- Separation between Product, Product Asset, Purchase, and Ownership.
- Preservation of asset identity and original first-purchase date across resale.
- The need for platform-origin and current-ownership eligibility checks for second-life listings.
- The separation between AI assistance and authoritative deterministic business logic.
- Protection of previous owners' personal information.

13.2 Items Requiring Team Validation

The following decisions remain to be confirmed unless already approved elsewhere in the project documentation:

- Final stakeholder and user-role list.
- Detailed role permissions and administrative responsibilities.
- Final acceptance thresholds for performance and availability.
- Exact behavior for disputes, returns, refunds, and transaction completion.
- Required evidence for each product category and the review process for that evidence.
- External service providers and integration contracts.
- Final technology stack and implementation architecture.
- Specific priority and delivery commitment for each requirement.

13.3 Approval Record

Role	Name	Status / Date
Team Representative	[TO BE COMPLETED]	[PENDING]
Team Members	[TO BE COMPLETED]	[PENDING]
Project Supervisor	[TO BE COMPLETED]	[PENDING]

This document becomes the approved requirements baseline only after the team and relevant reviewer validate its scope, assumptions, priorities, and open decisions.

14. Conclusion

This Requirements Specification establishes a structured foundation for TechWise by defining stakeholder needs, representative user interactions, functional capabilities, and non-functional quality expectations.

The requirements emphasize product information quality, explainable decision support, compatibility assessment, lifecycle continuity, ownership privacy, and evidence-aware second-life eligibility.

The central architectural principle is that AI supports understanding and explanation, while authoritative product records and deterministic backend logic govern facts, validation, authorization, calculations, compatibility, eligibility, and ownership state.

The next project activity is to validate these requirements and use them as input to System Analysis & Design, where the approved requirements can be translated into diagrams, data models, interfaces, and detailed technical specifications.